*.ops

Purpose

The six operation databases contain named parameter sets for management operations. A scheduled operation in management.sch or an action in a lum.dtl decision table names an entry, and SWAT+ uses that entry's values when the operation runs.

File	Contents	Used by
harv.ops	harvest operations	harv and hvkl in management.sch; harvest and harvest_kill in lum.dtl
graze.ops	grazing operations	graz in management.sch; graze in lum.dtl
irr.ops	irrigation operations	irrm, irrp and irpm in management.sch; irrigate and irr_demand in lum.dtl
chem_app.ops	application methods for fertilizer, manure and pesticide	fert, manu and pest in management.sch; fertilize, fert_future, manure, pest_apply and manure_demand in lum.dtl
fire.ops	burning operations	burn in management.sch and lum.dtl
sweep.ops	street sweeping operations	swep in management.sch

The six file names are given on the ops line of file.cio, in the order harvest, graze, irrigation, chemical application, fire, sweep. The defaults are the names in the table. If a file is missing, or named null in file.cio, the database is empty, and any operation that names an entry from it uses an entry with all values zero (irrigation depth 25.4 mm).

Format

All six files share one layout. Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one operation. The columns are read by position, so every column in the tables below must be present and in this order. A trailing description column, which the SWAT+ Editor writes, is ignored.

Names are matched exactly against the names used in management.sch and lum.dtl. A name that is not found leaves the operation pointing at an empty entry with zero values, and no message is written.

harv.ops

#	Column	Units	Typical range	Description
1	name	-	-	Harvest operation name.
2	harv_typ	-	see below	Harvest type.
3	harv_idx	fraction	0 to 1	Harvest index override: the fraction of above-ground biomass cut. Used by biomass harvests and, as a multiplier, by residue. Ignored by grain and root harvests.
4	harv_eff	fraction	0 to 1	Harvest efficiency: the fraction of the harvested material removed from the HRU.

#	Column	Units	Typical range	Description
5	`harv_bm_min`	kg/ha	0 and up	Minimum plant biomass for the harvest to take place. For `residue` harvests, the minimum surface residue left behind.

The harvest types and what each one does:

`harv_typ`	Harvest only (`harv`, `harvest`)	Harvest and kill (`hvkl`, `harvest_kill`)
`grain`	grain harvest	grain harvest
`picker`	grain harvest	grain harvest
`stripper`	biomass harvest	grain harvest
`biomass`	biomass harvest	biomass harvest
`tree`	biomass harvest	no harvest; the plant is only killed
`tuber`	root harvest	root harvest
`peanuts`	root harvest	root harvest
`residue`	residue harvest	residue harvest

A harvest of a growing plant runs only if the plant's total biomass is above `harv_bm_min`. A `residue` harvest scheduled after the plant has died runs without this check.

Grain harvest. The yield is $Y = \text{harv_eff} \cdot M_{seed}$, where M_{seed} is the seed (yield) pool built up during growth from the harvest index in `plants.plt`. The seed mass that is not removed goes to the top soil layer as organic matter. `harv_idx` has no effect.

Biomass harvest. A fraction $h = \text{harv_idx} \cdot \text{harv_eff}$ of the seed, leaf and stem mass is cut. The cut takes place only if the above-ground biomass left afterwards is above `harv_bm_min`. The cut mass is reported as yield, and a further $(1 - \text{harv_eff})$ times the cut mass is added to the surface residue. Leaf area and accumulated heat units are multiplied by $1 - 0.5\,h$.

Root harvest (`tuber`, `peanuts`). The yield is `harv_eff` times the root mass. The remaining roots stay in the soil as residue.

Residue harvest. Each plant's surface residue is reduced by the fraction $e \cdot \text{harv_idx}$, but not below `harv_bm_min`. The efficiency e is the `op_data3` value of the `management.sch` row (or `const` of the decision table action) when it is greater than zero, and `harv_eff` otherwise.

For every type, the reported yield is reduced by the plant's pest stress.

graze.ops

#	Column	Units	Typical range	Description
1	`name`	-	-	Grazing operation name.
2	`fert`	-	-	Name of the manure in `fertilizer.frt` that the animals deposit. Called `fertname` in earlier documentation.
3	`bm_eat`	kg/ha/day	0 to 500	Dry biomass removed by grazing each day.
4	`bm_tramp`	kg/ha/day	0 to 500	Dry biomass removed by trampling each day. Read but not applied.
5	`man_amt`	kg/ha/day	0 to 500	Dry manure deposited each day.
6	`grz_bm_min`	kg/ha	0 to 5000	Minimum above-ground biomass for grazing to take place on a given day.

A `graz` row in `management.sch` starts a grazing period, and its `op_data3` value is the length of the period in days. On each day of the period, if the HRU's above-ground biomass is at least `grz_bm_min`, the model removes `bm_eat`

from the plants (spread over the plants in proportion to their above-ground biomass, and never more than is there) and adds `man_amt` of the named manure to the top soil layer. While a plant's biomass is below 2500 kg/ha, grazing also lowers its leaf area index by `bm_eat`/1000 and multiplies its accumulated heat units by $1 - \text{bm_eat}/1000$. A `graze` action in a decision table applies one day of grazing each time it fires.

The parameter values in the SWAT+ Editor's default `graze.ops` are rough estimates. If grazing matters in a watershed, derive values from the stocking rate and type of animal. Trampling is often set equal to `bm_eat` (the animals trample as much as they eat), but in 62.0.1 `bm_tramp` has no effect.

irr.ops

#	Column	Units	Typical range	Description
1	`name`	-	-	Irrigation operation name.
2	`amt_mm`	mm	0 and up	Irrigation depth per application. Called `irr_amt` in earlier documentation.
3	`eff_frac`	fraction	0 to 1	In-field efficiency: the fraction of applied water available to the crop. Called `irr_eff` in earlier documentation.
4	`sumq_frac`	fraction	0 to 1	Fraction of the applied water that leaves as surface runoff. Called `surq_rto` in earlier documentation.
5	`dep_sub`	mm	0 and up	Application depth for subsurface irrigation. Read but not used. Called `irr_dep` in earlier documentation.
6	`salt_ppm`	ppm	0 and up	Salt concentration in the irrigation water. Used by the paddy irrigation operation `irrp`. Called `irr_salt` in earlier documentation.
7	`no3_ppm`	ppm	0 and up	Nitrate concentration in the irrigation water. Used by `irrp`. Called `irr_no3n` in earlier documentation.
8	`po4_ppm`	ppm	0 and up	Phosphate concentration in the irrigation water. Read but not used. Called `irr_po4` in earlier documentation.

A scheduled irrigation (`irrm` or `irpm` in `management.sch`) applies

$$I_{applied} = \text{amt_mm} \cdot \text{eff_frac} \cdot (1 - \text{sumq_frac}), \qquad I_{runoff} = \text{amt_mm} \cdot \text{sumq_frac}$$

The `irrigate` action in `lum.dtl` uses the action's `const` (mm) in place of `amt_mm` and takes `eff_frac` and `sumq_frac` from the entry named in its `option` column. The `irr_demand` action with file pointer `unlim` uses `amt_mm`, but computes runoff as $\text{amt_mm} \cdot \text{eff_frac} \cdot \text{sumq_frac}$. See `*.dtl` for both actions.

chem_app.ops

#	Column	Units	Typical range	Description
1	`name`	-	-	Application method name, for example `broadcast`, `band`, `foliar`, `inject`, `aerial_liquid`, `aerial_solid`, `drill`, `side_dress`, `fertigate`, `basal`, `rope_wick`, `tree_inject`.
2	`chem_form`	-	`solid`, `liquid`	Form of the chemical. Read but not used.

#	Column	Units	Typical range	Description
3	app_typ	-	spread, spray, inject, direct	Application type. Read but not used.
4	app_eff	fraction	0 to 1	Application efficiency. Multiplies the amount of a pest operation or pest_apply action. Not applied to fertilizer or manure.
5	foliar_eff	fraction	0 to 1	Foliar efficiency. Read but not used.
6	inject_dp	mm	0 and up	Injection depth. Read but not used.
7	surf_frac	fraction	0 to 1	Fraction of the applied fertilizer, manure or pesticide placed in the top 10 mm soil layer. The rest goes to the second soil layer.
8	drift_pot	fraction	0 to 1	Drift potential. Read but not used.
9	aerial_unif	fraction	0 to 1	Aerial uniformity. Read but not used.

A decision table fertilize action on an HRU with standing water applies the fertilizer to the water instead of the soil when the method's surf_frac is 1.

fire.ops

#	Column	Units	Typical range	Description
1	name	-	-	Fire operation name, for example grass, tree_intense, tree_low.
2	chg_cn2	-	0 to 20	Amount added to the HRU's curve number for moisture condition II. The result is capped at 98.
3	frac_burn	fraction	0 to 1	Fraction of above-ground plant biomass, surface residue and top-layer soil organic matter burned.

A burn removes frac_burn of the above-ground biomass of every plant, of the surface residue, and of the slow and passive humus in the top soil layer. The phosphorus in the burned material is returned to the passive humus pool, and the burned plant carbon is reported as emitted carbon.

sweep.ops

#	Column	Units	Typical range	Description
1	name	-	-	Street sweeping operation name.
2	swp_eff	fraction	0 to 1	Removal efficiency: the fraction of built-up solids removed from the swept curb length.
3	frac_curb	fraction	0 to 1	Fraction of the curb length that can be swept. Parked cars and other obstacles often make part of the curb unavailable.

Street sweeping is meant for urban HRUs that use the build-up and wash-off method. The removal efficiency depends on the type of sweeper, whether flushing is part of the cleaning, the amount of solids and the frequency of rain, and is usually higher for particulate constituents than for nitrogen and phosphorus (Pitt, 1979). SWAT+ assumes fixed nutrient concentrations in the solids, so one efficiency applies to every constituent.

In 62.0.1 a swep operation reads the entry and logs it, but the sweeping routine uses a curb fraction of zero and is never called during the run, so sweep.ops has no effect on results.

Example

`refdata/Osu_1hru` in the SWAT+ repository, first rows of each file:

```
harv.ops: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                      harv_typ      harv_idx      harv_eff   harv_bm_min  description
forest_cut                    tree       0.95000       0.99000       0.00000
grain                        grain       0.00000       0.95000       0.00000
grass_bag                  biomass       0.50000       1.00000    2000.00000
grass_mulch                biomass       0.50000       0.00000    2000.00000
```

`grass_bag` and `grass_mulch` both use a cut fraction of 0.5. With `harv_eff` at 1, `grass_bag` removes everything it cuts; with `harv_eff` at 0, `grass_mulch` cuts nothing ($h = 0.5 \times 0$). Both run only on plants above 2000 kg/ha. `grain` removes 95% of the seed pool.

```
graze.ops: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                          fert        bm_eat      bm_tramp       man_amt    grz_bm_min  description
dairy_high                dairy_fr      25.00000      10.00000       0.95000    3000.00000  High_Prod_Dair
```

```
irr.ops: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                       amt_mm      eff_frac     sumq_frac       dep_sub      salt_ppm       no3_ppm
sprinkler_med          25.00000       0.85000       0.00000       0.00000     100.00000      20.00000
drip                   25.00000       0.90000       0.00000     100.00000     100.00000      20.00000
```

A scheduled `sprinkler_med` irrigation puts $25 \times 0.85 = 21.25$ mm of water into the soil.

```
chem_app.ops: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                     chem_form           app_typ       app_eff    foliar_eff     inject_dp     surf_fr
broadcast                    solid            spread       0.90000       0.00000       0.00000       1.000
inject                      liquid            inject       0.95000       0.00000     150.00000       0.200
```

```
fire.ops: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                     chg_cn2     frac_burn  description
grass                    8.00000       1.00000
tree_low                 6.00000       0.70000
```

Related files

- `management.sch`: scheduled operations that name entries in these files.
- `*.dtl`: decision table actions that name entries in these files.
- `fertilizer.frt`: the manure named in `graze.ops`, and the fertilizers applied with a `chem_app.ops` method.
- `plants.plt`: the plant harvest index that builds the seed pool used by grain harvests.
- `file.cio`: names the six files on its `ops` line.

Sources

- Readers: `mgt_read_harvops.f90`, `mgt_read_grazeops.f90`, `mgt_read_irrops.f90`, `mgt_read_chemapp.f90`, `mgt_read_fireops.f90`, `mgt_read_sweepops.f90`, called from `proc_db.f90`
- Type definitions: `mgt_operations_module.f90`
- File name slots: `type input_ops` in `input_file_module.f90`
- Scheduled operations: `mgt_sched.f90`; decision table actions: `actions.f90`
- Harvest: `mgt_harvgrain.f90`, `mgt_harvbiomass.f90`, `mgt_harvtuber.f90`, `mgt_harvresidue.f90`
- Grazing: `pl_graze.f90`, daily loop in `hru_control.f90`
- Burning: `pl_burnop.f90`
- Street sweeping: `hru_sweep.f90`, `hru_urbanhr.f90`
- Application methods: `pl_fert.f90`, `pest_apply.f90`

File	Column	Fortran variable
harv.ops	harv_typ	harvop_db%typ
harv.ops	harv_idx	harvop_db%hi_ovr
harv.ops	harv_eff	harvop_db%eff

File	Column	Fortran variable
harv.ops	harv_bm_min	harvop_db%bm_min
graze.ops	fert	grazeop_db%fertnm (index stored in grazeop_db%manure_id)
graze.ops	bm_eat	grazeop_db%eat
graze.ops	bm_tramp	grazeop_db%tramp
graze.ops	man_amt	grazeop_db%manure
graze.ops	grz_bm_min	grazeop_db%biomin
irr.ops	eff_frac	irrop_db%eff
irr.ops	sumq_frac	irrop_db%surq
irr.ops	dep_sub	irrop_db%dep_mm
irr.ops	salt_ppm, no3_ppm, po4_ppm	irrop_db%salt, %no3, %po4
chem_app.ops	chem_form	chemapp_db%form
chem_app.ops	app_typ	chemapp_db%op_typ
chem_app.ops	inject_dp	chemapp_db%inject_dep
fire.ops	chg_cn2	fire_db%cn2_upd
fire.ops	frac_burn	fire_db%fr_burn
sweep.ops	swp_eff	sweepop_db%eff
sweep.ops	frac_curb	sweepop_db%fr_curb